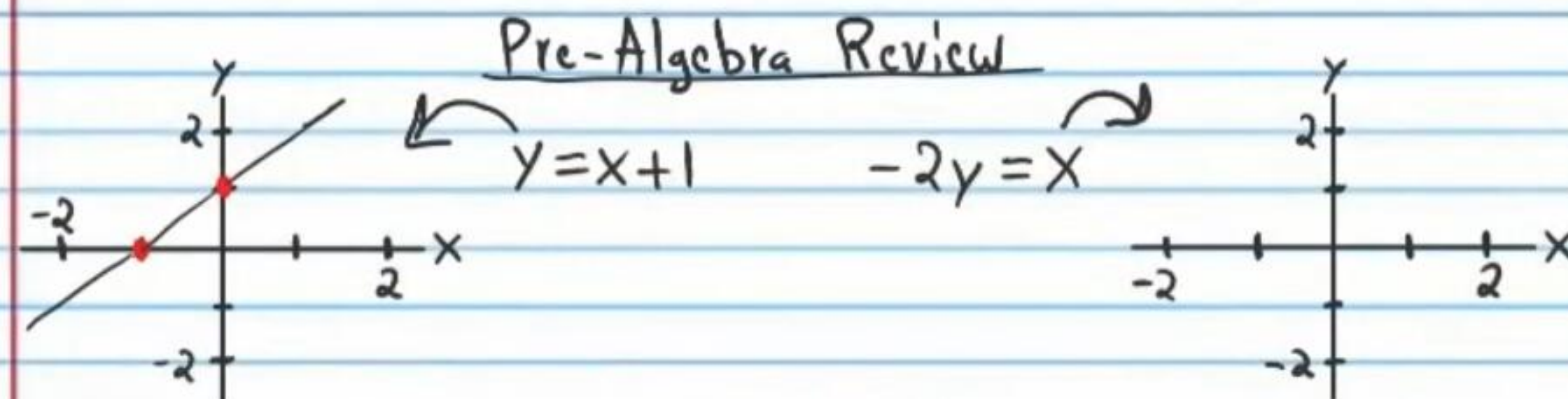


The Three Forms of Linear Equations and Finding the Slopes of Lines



The Slope of a Line

If (X_1, Y_1) and (X_2, Y_2) are two points on a line, the slope of the line is defined by the equation below.

$$m = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Slope measures the steepness of a line. If the slope of a line is positive, the line travels up moving from left to right. If the slope of a line is negative, the line travels down moving left to right. If the slope of a line is zero, the line is horizontal. If the slope of a line is undefined, the line is vertical.

The Three Forms of a Linear Equation

<u>Standard Form</u>	<u>Slope-Intercept Form</u>	<u>Point-Slope Form</u>
$AX + BY = C$	$y = mx + b$	$y - y_p = m(x - x_p)$
A, B, and C are constants and X and Y are variables.	m = Slope, b = y-intercept, and x and y are variables.	m = slope, x_p is the X-coordinate of a known point, and y_p is the Y-coordinate of the same known point. X and Y are variables.

Identify the form of each linear equation below by writing "S" for standard, "SI" for slope-intercept, or "PS" for point-slope.

① $2x + 4y = 3$ ② $y = \frac{5}{6}x + 3$ ③ $4 = -8x + 7y$

S SI S

④ $y - 4 = -3(x - 5)$ ⑤ $2x + y = -5$ ⑥ $y = 2x + \frac{4}{3}$

PS S SI

⑦ $-4x - 2y = 5$ ⑧ $y + 1 = -\frac{3}{10}(x + 1)$ ⑨ $\frac{1}{4}x - \frac{2}{9}y = -\frac{1}{3}$

⑩ $-3.98 = -0.4x + 2.7y$ ⑪ $-\frac{7}{4}x + 1 = y$ ⑫ $3x - 6y = 6$

⑬ $y = -\frac{2x}{3} - \frac{7}{8}$ ⑭ $1.3x + 6 = y$ ⑮ $\frac{4}{5}(x - 7) = y + 4$

⑯ $3x + 5y = 2$ ⑰ $y = \frac{2}{5}x + 7$ ⑱ $4 = -9x + 2y$

S SI S

⑲ $-8(x - 9) = y - 2$ ⑳ $4x + 3y = -6$ ㉑ $y = 4x + \frac{8}{9}$

PS S SI

㉒ $-3x - 2y = 7$ ㉓ $y + 7 = -\frac{3}{8}(x + 6)$ ㉔ $\frac{3}{4}x - \frac{7}{9}y = -\frac{1}{2}$

S PS S

$$(25) -1.99 = -0.3x + 5.7y \quad (26) -\frac{5}{4}x + 2 = y \quad (27) 6x - 7y = 3$$

$\boxed{S}$

$\boxed{SI}$

$\boxed{S}$

$$(28) y = -\frac{5x}{3} - \frac{7}{10} \quad (29) 1.7x + 8 = y \quad (30) \frac{4}{7}(x-8) = y+5$$

$\boxed{SI}$

$\boxed{SI}$

$\boxed{PS}$

Identify the slope and y-intercept of each of the following linear equations.

$$(31) y = \frac{4}{3}x + 1 \quad (32) y = 2x + \frac{3}{8} \quad (33) -\frac{2}{7}x + \frac{5}{6} = y$$

$$\boxed{m = \frac{4}{3}} \quad \boxed{b = 1}$$

$$\boxed{m = 2} \quad \boxed{b = \frac{3}{8}}$$

$$\boxed{m = -\frac{2}{7}} \quad \boxed{b = \frac{5}{6}}$$

$$(34) y = \frac{x}{2} - 4 \quad (35) y = \frac{4}{9}x$$

$$\boxed{m = \frac{1}{2}} \quad \boxed{b = -4}$$

$$\boxed{m = \frac{4}{9}} \quad \boxed{b = 0}$$

$$(36) -4x - 8 = y$$

$$\boxed{m = -4} \quad \boxed{b = -8}$$

$$(37) y = -\frac{6}{11}x \quad (38) -\frac{1}{5}x + \frac{1}{2} = y$$

$$\boxed{m = -\frac{6}{11}} \quad \boxed{b = 0}$$

$$(39) y = \frac{x}{4}$$

$$(40) y = x - 1 \quad (41) y = \frac{2x}{3} + 5 \quad (42) y = -\frac{x}{10} - \frac{3}{5}$$

$$(43) y = \frac{4}{3}x + 2 \quad (44) y = 3x + \frac{9}{11} \quad (45) -\frac{6}{7}x + \frac{5}{8} = y$$

$$\boxed{m = \frac{4}{3}} \quad \boxed{b = 2}$$

$$\boxed{m = 3} \quad \boxed{b = \frac{9}{11}}$$

$$\boxed{m = -\frac{6}{7}} \quad \boxed{b = \frac{5}{8}}$$

$$(46) y = \frac{9}{2}x - 4 \quad (47) y = \frac{4}{5}x$$

$$\boxed{m = \frac{9}{2}} \quad \boxed{b = -4}$$

$$\boxed{m = \frac{4}{5}} \quad \boxed{b = 0}$$

$$(48) -2x - 3 = y$$

$$\boxed{m = -2} \quad \boxed{b = -3}$$

$$(49) y = -\frac{6}{13}x \quad (50) -\frac{4}{7}x + \frac{3}{2} = y \quad (51) \frac{x}{3} = y$$

$$\boxed{m = -\frac{6}{13}} \quad \boxed{b = 0}$$

$$\boxed{m = -\frac{4}{7}} \quad \boxed{b = \frac{3}{2}}$$

$$\boxed{m = \frac{1}{3}} \quad \boxed{b = 0}$$

$$(52) y = 3x - 1$$

$$\boxed{m = 3} \quad \boxed{b = -1}$$

$$(53) y = \frac{3x}{7} + 9$$

$$\boxed{m = \frac{3}{7}} \quad \boxed{b = 9}$$

$$(54) y = -\frac{2x}{5} - 3$$

$$\boxed{m = -\frac{2}{5}} \quad \boxed{b = -3}$$

Write linear equations with the following slopes and y-intercepts.

$$(55) m = 2, b = \frac{1}{3} \quad (56) m = -\frac{2}{5}, b = 4 \quad (57) m = -7, b = -3$$

$$\boxed{y = 2x + \frac{1}{3}}$$

$$\boxed{y = -\frac{2}{5}x + 4}$$

$$\boxed{y = -7x - 3}$$

$$(58) m = \frac{3}{4}, b = -\frac{8}{9} \quad (59) m = \frac{1}{6}, b = 2 \quad (60) m = 5, b = -3.7$$

$$\boxed{y = \frac{3}{4}x - \frac{8}{9}}$$

$$\boxed{y = \frac{1}{6}x + 2}$$

$$\boxed{y = 5x - 3.7}$$

$$(61) m = \frac{11}{5}, b = -1 \quad (62) m = 1, b = 1 \quad (63) m = 3, b = \frac{2}{7}$$

$$\boxed{y = \frac{11}{5}x - 1}$$

$$\boxed{y = x + 1}$$

$$\boxed{y = 3x + \frac{2}{7}}$$

$$(64) m = -\frac{3}{8}, b = 2 \quad (65) m = -5, b = -3 \quad (66) m = \frac{5}{2}, b = -\frac{2}{3}$$

$$\boxed{y = -\frac{3}{8}x + 2}$$

$$\boxed{y = -5x - 3}$$

$$\boxed{y = \frac{5}{2}x - \frac{2}{3}}$$

$$(67) m = \frac{1}{4}, b = 5 \quad (68) m = 2, b = -3.9 \quad (69) m = \frac{7}{5}, b = 0$$

$$\boxed{y = \frac{1}{4}x + 5}$$

$$\boxed{y = 2x - 3.9}$$

$$\boxed{y = \frac{7}{5}x}$$

$$(70) m = -1, b = \frac{3}{7}$$

$$\boxed{y = -x + \frac{3}{7}}$$

Find the slope of each of the following linear equations and a point on the line produced by graphing the equation.

71) $y - 4 = \frac{2}{7}(x - 5)$ 72) $y - 7 = -\frac{1}{3}(x + 9)$ 73) $5(x - 2) = y + 3$

$m = \frac{2}{7}$ $(5, 4)$ $m = -\frac{1}{3}$ $(-9, 7)$ $m = 5$ $(2, -3)$

74) $\frac{5}{12}(x + 4) = y + 3$ 75) $y - 2 = \frac{2}{3}(x - 5)$ 76) $y - 7 = -\frac{5}{6}(x + 1)$

$m = \frac{5}{12}$ $(-4, -3)$ $m = \frac{2}{3}$ $(5, 2)$ $m = -\frac{5}{6}$ $(-1, 7)$

77) $y + 3 = 2(x - 8)$ 78) $\frac{1}{5}(x + 4) = y + 2$ 79) $y = -2(x + 4)$

$m = 2$ $(8, -3)$ $m = \frac{1}{5}$ $(-4, -2)$ $m = -2$ $(-4, 0)$

In each problem below, write an equation of the line that contains the given point and has the given slope.

80) $(6, 8)$, $m = 4$ 81) $(-5, 3)$, $m = \frac{6}{11}$ 82) $(2, -3)$, $m = -6$

$y - 8 = 4(x - 6)$ $y - 3 = \frac{6}{11}(x + 5)$ $y + 3 = -6(x - 2)$

83) $(-1, -3)$, $m = \frac{1}{6}$ 84) $(2, -9)$, $m = -\frac{3}{5}$ 85) $(7, 7)$, $m = -1$

$y + 3 = \frac{1}{6}(x + 1)$ $y + 9 = -\frac{3}{5}(x - 2)$ $y - 7 = -(x - 7)$

86) $(-4, -5)$, $m = -\frac{10}{3}$ 87) $(3.7, -\frac{2}{9})$, $m = 6.5$ 88) $(5, 3)$, $m = 2$

$y + 5 = -\frac{10}{3}(x + 4)$ $y + \frac{2}{9} = 6.5(x - 3.7)$ $y - 3 = 2(x - 5)$

89) $(-8, 2)$, $m = \frac{3}{5}$ 90) $(2, -7)$, $m = -4$ 91) $(-3, -6)$, $m = \frac{7}{6}$

$y - 2 = \frac{3}{5}(x + 8)$ $y + 7 = -4(x - 2)$ $y + 6 = \frac{7}{6}(x + 3)$

92) $(1, -6)$, $m = -\frac{2}{3}$ 93) $(4, 9)$, $m = -2$ 94) $(-3, -2)$, $m = -\frac{5}{3}$

$y + 6 = -\frac{2}{3}(x - 1)$ $y - 9 = -2(x - 4)$ $y + 2 = -\frac{5}{3}(x + 3)$

95) $(-\frac{2}{3}, 2.6)$, $m = \frac{3}{8}$ 96) $(0, 3)$, $m = 3$

$y - 2.6 = \frac{3}{8}(x + \frac{2}{3})$ $y - 3 = 3(x - 0)$
 $y - 3 = 3x$

A) Convert the following linear equations from standard form to slope-intercept form.

B) Then find the slope and y-intercept of the lines produced by graphing the equations.

96) $4x + 2y = 8$ 97) $18 = -6x + 3y$ 98) $20x - 5y = 25$

$2y = -4x + 8$ $6x + 18 = 3y$ $-5y = -20x + 25$

$\frac{2y}{2} = \frac{-4x + 8}{2}$ $\frac{6x + 18}{3} = \frac{3y}{3}$ $\frac{-5y}{-5} = \frac{-20x + 25}{-5}$

A) $y = -2x + 4$ A) $2x + 6 = y$ A) $y = 4x - 5$

B) $m = -2$ $b = 4$ B) $m = 2$ $b = 6$ B) $m = 4$ $b = -5$

99) $3 = 7x + 4y$ 100) $3x + 2y = -4$ 101) $2x - 6y = 8$

$-7x + 3 = 4y$ $2y = -3x - 4$ $-6y = -2x + 8$

$\frac{-7x + 3}{4} = \frac{4y}{4}$ $\frac{2y}{2} = \frac{-3x - 4}{2}$ $y = \frac{-2x + 8}{-6}$

A) $-\frac{7}{4}x + \frac{3}{4} = y$ A) $y = -\frac{3x}{2} - 2$ A) $y = -\frac{2x}{-6} + \frac{8}{-6}$

B) $m = -\frac{7}{4}$ $b = \frac{3}{4}$ B) $m = -\frac{3}{2}$ $b = -2$ B) $m = \frac{1}{3}$ $b = -\frac{4}{3}$

⑩② $2 = x - 4y$ ⑩③ $-x + y = -6$ ⑩④ $-5x + 20y = -15$

⑩⑤ $-15x - 9y = 45$ ⑩⑥ $8x + 2y = 16$ ⑩⑦ $-7x + 7y = 21$

$2y = -8x + 16$ $7y = 7x + 21$
 A) $y = -4x + 8$ A) $y = x + 3$
 B) $m = -4$ $b = 8$ B) $m = 1$ $b = 3$

⑩⑧ $15x - 5y = 10$ ⑩⑨ $2x + 3y = 5$ ⑩⑩ $5x + 3y = -2$

$-5y = -15x + 10$ $3y = -2x + 5$ $3y = -5x - 2$
 A) $y = 3x - 2$ A) $y = -\frac{2}{3}x + \frac{5}{3}$ A) $y = -\frac{5}{3}x - \frac{2}{3}$
 B) $m = 3$ $b = -2$ B) $m = -\frac{2}{3}$ $b = \frac{5}{3}$ B) $m = -\frac{5}{3}$ $b = -\frac{2}{3}$

⑩⑪ $3x - 7y = -5$ ⑩⑫ $3 = -x + 4y$ ⑩⑬ $2x + y = -9$

$-7y = -3x - 5$ $x + 3 = 4y$
 A) $y = \frac{3}{7}x + \frac{5}{7}$ A) $\frac{x}{4} + \frac{3}{4} = y$ A) $y = -2x - 9$
 B) $m = \frac{3}{7}$ $b = \frac{5}{7}$ B) $m = \frac{1}{4}$ $b = \frac{3}{4}$ B) $m = -2$ $b = -9$

⑩⑭ $-3x + 18y = -8$

$18y = 3x - 8$
 $y = \frac{3x}{18} - \frac{8}{18}$
 A) $y = \frac{x}{6} - \frac{4}{9}$
 B) $m = \frac{1}{6}$ $b = -\frac{4}{9}$

⑩⑮ $-16x - 9y = 6$

$-9y = 16x + 6$
 $y = -\frac{16x}{9} - \frac{6}{9}$
 A) $y = -\frac{16}{9}x - \frac{2}{3}$
 B) $m = -\frac{16}{9}$ $b = -\frac{2}{3}$

A) Convert the following equations from point-slope form to slope-intercept form. B) Then find the slope and y-intercept.

⑩⑯ $y + 3 = 4(x - 2)$ ⑩⑰ $y - 4 = -2(x + 7)$ ⑩⑱ $x + 7 = y + 4$

$y + 3 = 4x - 8$ $y - 4 = -2x - 14$ A) $x + 3 = y$
 A) $y = 4x - 11$ A) $y = -2x - 10$ B) $m = 1$ $b = 3$
 B) $m = 4$ $b = -11$ B) $m = -2$ $b = -10$

$$\textcircled{119} \frac{2}{3}(x-2)=y-4 \quad \textcircled{120} y+6=4(x-2) \quad \textcircled{121} y-4=-3(x+3)$$

$$\frac{2}{3}x - \frac{4}{3} = y - 4$$

$$y + 6 = 4x - 8$$

$$y - 4 = -3x - 9$$

$$\frac{2}{3}x - \frac{4}{3} + \frac{12}{3} = y$$

$$A) \boxed{y = 4x - 14}$$

$$A) \boxed{y = -3x - 5}$$

$$A) \boxed{\frac{2}{3}x + \frac{8}{3} = y}$$

$$B) \boxed{m = 4} \quad \boxed{b = -14}$$

$$B) \boxed{m = -3} \quad \boxed{b = -5}$$

$$D) \boxed{m = \frac{2}{3}} \quad \boxed{b = \frac{8}{3}}$$

$$\textcircled{122} y + 2 = x + 3$$

$$A) \boxed{y = x + 1}$$

$$B) \boxed{m = 1} \quad \boxed{b = 1}$$

$$\textcircled{123} \frac{2}{5}(x-3)=y-9$$

$$\frac{2}{5}x - \frac{6}{5} = y - 9$$

$$B) \boxed{m = \frac{2}{5}}$$

$$A) \boxed{\frac{2}{5}x + \frac{39}{5} = y}$$

$$\boxed{b = \frac{39}{5}}$$

In each problem below, use the slope equation to find the slope of the line that passes through the two given points.

$$\textcircled{124} (4,3) \text{ \& } (2,6) \quad \textcircled{125} (2,-3) \text{ \& } (5,7) \quad \textcircled{126} (-18,4) \text{ \& } (2,-4)$$

$$\begin{matrix} \downarrow & \downarrow \\ (x_1, y_1) & (x_2, y_2) \end{matrix}$$

$$m = \frac{6-3}{2-4} = \boxed{-\frac{3}{2}}$$

$$m = \frac{7-(-3)}{5-2} = \boxed{\frac{10}{3}}$$

$$m = \frac{-4-4}{2-(-18)} = \frac{-8}{2+18} = \frac{-8}{20} = \boxed{-\frac{2}{5}}$$

$$\textcircled{127} (-8,-6) \text{ \& } (-7,-9) \quad \textcircled{128} (4,3) \text{ \& } (5,6) \quad \textcircled{129} (2,6) \text{ \& } (1,4)$$

$$m = \frac{4-6}{1-2} = \frac{-2}{-1} = \boxed{2}$$

$$m = \frac{-2}{-1} = \boxed{2}$$

$$\textcircled{130} (5,-3) \text{ \& } (6,7)$$

$$m = \frac{7-(-3)}{6-5} = \frac{10}{1} = \boxed{10}$$

$$m = \frac{7+3}{1} = \boxed{10}$$

$$\textcircled{133} (1,1) \text{ \& } (4,2)$$

$$\textcircled{131} (-9,4) \text{ \& } (3,-4)$$

$$m = \frac{-4-4}{3-(-9)} = \frac{-8}{12} = \boxed{-\frac{2}{3}}$$

$$m = \frac{-8}{12} = \boxed{-\frac{2}{3}}$$

$$\textcircled{132} (-9,-3) \text{ \& } (-7,-5)$$

$$m = \frac{-5-(-3)}{-7-(-9)} = \frac{-5+3}{-7+9} = \frac{-2}{2} = \boxed{-1}$$

$$m = \frac{-5+3}{-7+9} = \frac{-2}{2} = \boxed{-1}$$

$$m = \frac{-2}{2} = \boxed{-1}$$

$$\boxed{m = -1}$$

A) In each problem below, use the slope equation to find the slope of the line that contains the two given points.
B) Then write the point-slope form of the equation of the line containing the two given points.
C) Then convert the equations from point-slope form to slope-intercept form.

$$\textcircled{134} (4,7) \text{ \& } (3,5)$$

$$A) m = \frac{5-7}{3-4} = \frac{-2}{-1} = \boxed{2}$$

$$m = \frac{-2}{-1} = \boxed{2}$$

$$\boxed{m = 2}$$

$$B) \boxed{y - 7 = 2(x - 4)}$$

$$C) y - 7 = 2x - 8$$

$$\boxed{y = 2x - 1}$$

$$\textcircled{135} (6,-2) \text{ \& } (1,8)$$

$$A) m = \frac{8-(-2)}{1-6} = \frac{10}{-5} = \boxed{-2}$$

$$m = \frac{10}{-5} = \boxed{-2}$$

$$B) \boxed{y + 2 = -2(x - 6)}$$

$$C) y + 2 = -2x + 12$$

$$\boxed{y = -2x + 10}$$

$$\textcircled{136} (7,11) \text{ \& } (3,-5)$$

$$A) m = \frac{-5-11}{3-7} = \frac{-16}{-4} = \boxed{4}$$

$$= \frac{-16}{-4} = \boxed{4}$$

$$B) \boxed{y - 11 = 4(x - 7)}$$

$$C) y - 11 = 4x - 28$$

$$\boxed{y = 4x - 17}$$

- A) slope
B) PS Form
C) convert to SI

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

137) (6,5) & (2,17) 138) (1,-2) & (-5,3) 139) (3,4) & (5,12)

A) $m = \frac{17-5}{2-6}$

$m = \frac{12}{-4}$

$m = \boxed{-3}$

B) $y-5 = -3(x-6)$

C) $y-5 = -3x+18$

$y = -3x+23$

A) $m = \frac{3-(-2)}{-5-1}$

$m = \frac{5}{-6}$

B) $y+2 = -\frac{5}{6}(x-1)$

C) $y+2 = -\frac{5}{6}x + \frac{5}{6}$

$y = -\frac{5}{6}x - \frac{7}{6}$

A) $m = \frac{12-4}{5-3}$

$m = \frac{8}{2}$

$m = \boxed{4}$

B) $y-4 = 4(x-3)$

C) $y-4 = 4x-12$

$y = 4x-8$

140) (4,5) & (-3,7) 141) (5,5) & (6,10) 142) (3,9) & (-2,0)

A) $m = \frac{7-5}{-3-4}$

$m = \boxed{-\frac{2}{7}}$

B) $y-5 = -\frac{2}{7}(x-4)$

C) $y-5 = -\frac{2}{7}x + \frac{8}{7}$

$y = -\frac{2}{7}x + \frac{13}{7}$

A) $m = \frac{10-5}{6-5}$

$m = \frac{5}{1}$

$m = \boxed{5}$

B) $y-5 = 5(x-5)$

C) $y-5 = 5x-25$

$y = 5x-20$

143) (2,3) & (-3,-5) 144) (8,2) & (7,-2)

Use the slope-intercept equation and the given information to find the slope-intercept form of the equation of the line.

145) $m = 6, (2,-3)$

$y = mx+b$

$-3 = 6(2)+b$

$-3 = 12+b$

$-15 = b$

$y = mx+b$

$y = 6x-15$

146) $b = -1, (-3,4)$

$y = mx+b$

$4 = m(-3) - 1$

$4 = -3m - 1$

$5 = -3m$

$-\frac{5}{3} = m$

$y = mx+b$

$y = -\frac{5}{3}x - 1$

147) $m = -2, (-4,-4)$

$y = mx+b$

$-4 = -2(-4)+b$

$-4 = 8+b$

$-12 = b$

$y = mx+b$

$y = -2x-12$

$$\textcircled{148} \ b = -8, (-2, -4) \quad \textcircled{149} \ m = 7, (-3, 1) \quad \textcircled{150} \ b = 3, (5, -5)$$

$$y = mx + b$$

$$y = mx + b$$

$$y = mx + b$$

$$-4 = m(-2) - 8$$

$$1 = 7(-3) + b$$

$$-5 = m(5) + 3$$

$$-4 = -2m - 8$$

$$1 = -21 + b$$

$$-5 = 5m + 3$$

$$-2 = m$$

$$22 = b$$

$$-\frac{8}{5} = m$$

$$y = mx + b$$

$$y = mx + b$$

$$y = mx + b$$

$$\boxed{y = -2x - 8}$$

$$\boxed{y = 7x + 22}$$

$$\boxed{y = -\frac{8}{5}x + 3}$$

$$\textcircled{151} \ m = -2, (5, -6) \quad \textcircled{152} \ b = 2, (-3, 4) \quad \textcircled{153} \ m = -10, (-3, -9)$$

$$y = mx + b$$

$$y = mx + b$$

$$-6 = -2(5) + b$$

$$4 = m(-3) + 2$$

$$-6 = -10 + b$$

$$4 = -3m + 2$$

$$4 = b$$

$$-\frac{2}{3} = m$$

$$y = mx + b$$

$$y = mx + b$$

$$\boxed{y = -2x + 4}$$

$$y = mx + b$$

$$\boxed{y = -\frac{2}{3}x + 2}$$

$$\textcircled{154} \ m = -\frac{3}{4}, (-3, \frac{1}{2}) \quad \textcircled{155} \ b = \frac{1}{3}, (-1, 0) \quad \textcircled{156} \ m = -\frac{8}{9}, (-2, 1)$$